

Data Fabric

The *data fabric* architecture is an evolution of the modern data warehouse (MDW) architecture: an advanced layer built onto the MDW to enhance data accessibility, security, discoverability, and availability. Picture the data fabric weaving its way throughout your entire company, accumulating all of the data and providing it to everyone who needs it, within your company or even outside it. It's an architecture that can consume data no matter the size, speed, or type. The most important aspect of the data fabric philosophy is that a data fabric solution can consume any and all data within the organization.

That is *my* definition of a data fabric; others in the industry define it differently. Some even use the term interchangeably with *modern data warehouse*! For instance, the consulting firm Gartner starts with a **similar definition**, writing that the data fabric is

a design concept that serves as an integrated layer (fabric) of data and connecting processes. A data fabric utilizes continuous analytics over existing, discoverable and inferred metadata assets to support the design, deployment and utilization of integrated and reusable data across all environments, including hybrid and multi-cloud platforms.

Where Gartner's view diverges from mine is in the view that data virtualization (which you learned about in **Chapter 6**) is a major piece of the data fabric technology that reduces the need to move or copy data from siloed systems. Gartner envisions an "intelligent data fabric" that leverages knowledge graphs and AI and ML for automating data discovery and cataloging, finding data relationships, orchestrating and integrating data, and auto-discovering metadata. However, much of the needed technology does not yet exist, and so far, neither does a data fabric that would satisfy Gartner's definition. I'm hopeful that technology will be available one day, but for now I recommend focusing on data fabric architectures as they exist today.

Data fabric is the most widely used architecture for new solutions being built in the cloud, especially for large amounts of data (over 10 terabytes). A lot of companies are “upgrading” their MDW architecture into a data fabric by adding additional technology, especially real-time processing, data access policies, and a metadata catalog. This chapter outlines eight technology features that take an MDW into data fabric territory. There’s some ambiguity here; if you add three of those features, do you have an MDW or a data fabric? There is no bright line between the two. Personally, I call it a data fabric architecture if it uses at least three of those eight technologies. Many people use none of them but nonetheless call their solution a data fabric because they abide by the data fabric philosophy (I think this is perfectly fine).

This chapter gives you an overview of the data fabric architecture and its eight key components: data access policies, a metadata catalog, master data management (MDM), data virtualization, real-time processing, APIs, services, and products.

The Data Fabric Architecture

The data fabric architecture contains all of the components of the MDW but adds several features. [Figure 11-1](#) shows a diagram of this architecture, with the stages of the data’s journey labeled by number like they were in [Figure 10-2](#). Data in a data fabric follows the same process as in an MDW; as you may recall from [Chapter 10](#), those stages are (1) ingestion, (2) storage, (3) transformation, (4) modeling, and (5) visualization.

So what does a data fabric offer above and beyond the MDW? This section will take a look at its added features.

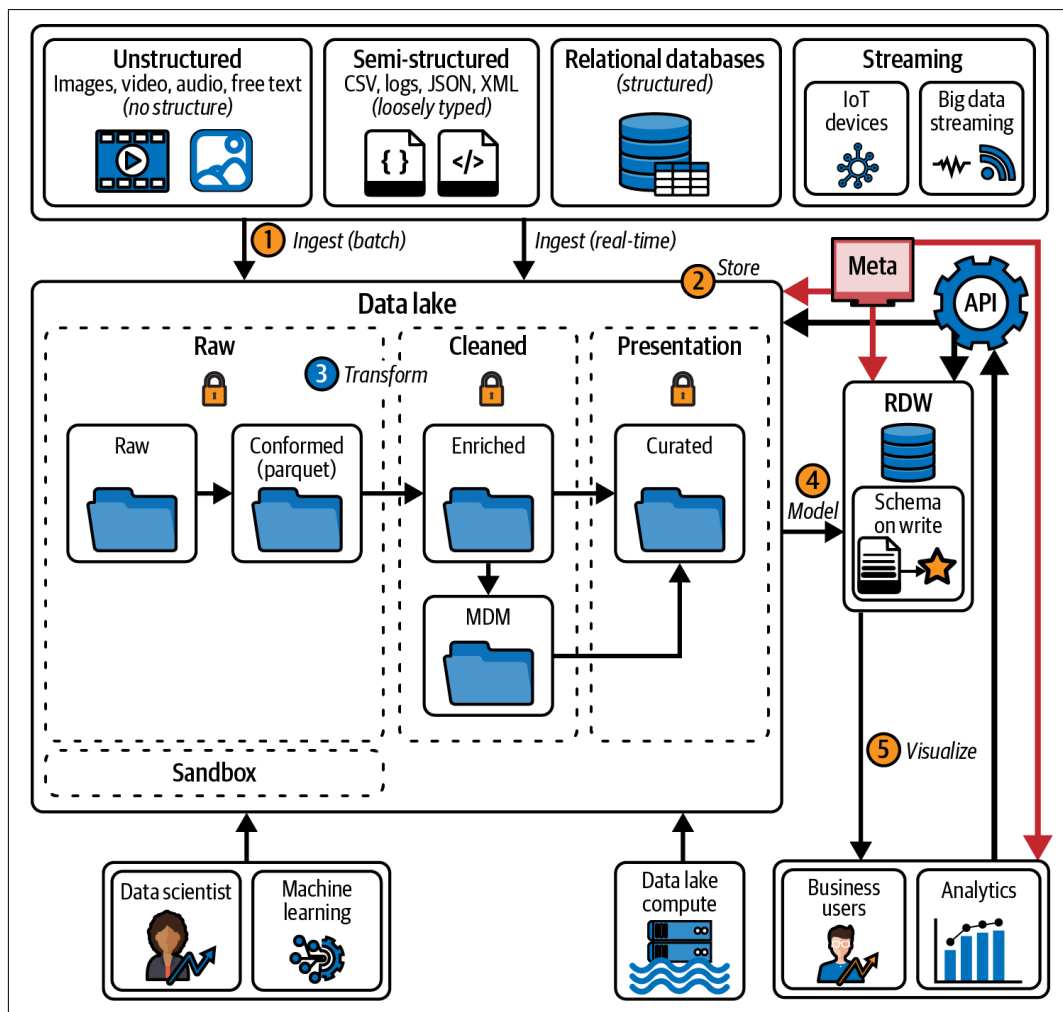


Figure 11-1. An overview of the journey data takes through a data fabric architecture

Data Access Policies

Data access policies are the key to data governance. They comprise a set of guidelines, rules, and procedures that control who has access to what information, how that information may be used, and when access can be granted or denied within an organization. They help ensure the security, privacy, and integrity of sensitive data, as well as compliance with laws and regulations, such as the **UK's General Data Protection Regulation (GDPR)** and the **US's Health Insurance Portability and Accountability Act (HIPAA)**. These policies typically cover topics like data classification, user authentication and authorization, data encryption, data retention, data backup and recovery, and data disposal.

All data requests within an organization must adhere to its established data access policies and must be processed through specific mechanisms, such as APIs, drivers, or data virtualization, ensuring compliance with regulations such as HIPAA. For instance, a healthcare organization's data access policies might require that only authorized medical staff have access to patient medical records. These policies should outline procedures for verifying staff credentials and what uses of the information are acceptable. To facilitate this controlled access, the organization might implement APIs that interface with secure authentication systems. With this security in place, when a health care provider requests a patient's record, the API checks the provider's credentials against the access policies before granting access.

Metadata Catalog

Data fabrics include a *metadata catalog*: a repository that stores information about data assets, including their structure, relationships, and characteristics. It provides a centralized, organized way to manage and discover data, making it easier for users to find and understand the data they need. For example, say a user searches the metadata catalog for "customer." The results will include any files, database tables, reports, or dashboards that contain customer data. Making it possible to see what ingestion and reporting has already been done helps everyone avoid duplicate efforts.

An important part of the catalog is *data lineage*, a record of the history of any given piece of data, including where it came from, how it was transformed, and where it is stored. Data lineage is used to comply with regulations such as data privacy laws. It's also where you can trace the origin of data and the transformations it has undergone, which is an important part of understanding how it was created and how reliable it is. You wouldn't want to base important decisions on data unless you knew you could trust it! For example, if a user asks about a particular value in a report, you can look up its lineage in the metadata catalog to show how that value came about.

Master Data Management

As you learned in [Chapter 6](#), MDM is the process of collecting, consolidating, and maintaining consistent and accurate data from various sources within a company to create a single authoritative source for master data. This data is typically nontransactional and describes the key entities of a company, such as customers, products, suppliers, and employees.

MDM helps companies make informed decisions and avoid data-related problems such as duplicate records, inconsistencies, and incorrect information.

Data Virtualization

You also encountered data virtualization in [Chapter 6](#). To refresh your memory, data virtualization is a software architecture that allows applications and end users to access data from multiple sources as if it were stored in a single location. The virtualized data is stored in a logical layer that acts as an intermediary between the end users and the data sources. That virtual layer is a single point of access that abstracts the complexity of the underlying data sources, making it easy for end users to access and combine data from multiple sources in real time without needing to copy or physically integrate it.

Using data virtualization does not mean you need to connect *all* data sources. In most cases, it's used only for a small number of situations, with most data still being centralized. Some in the industry see data virtualization as a definitive feature of a data fabric, arguing that an architecture is not a data fabric without virtualization; I disagree. In my definition of data fabric, virtualization is optional.

Real-Time Processing

As you saw in [Chapter 9](#), *real-time processing* refers to processing data and producing immediate results as soon as that data becomes available, without any noticeable delay. This allows you to make decisions based on up-to-date information (for example, traffic information while driving).

APIs

Instead of relying on connection strings, APIs provide data in a standardized way from a variety of sources—such as a data lake or the RDW—without sharing the particulars of where the data is located. If the data is moved, you only need to update the API's internal code; none of the applications calling the API are affected. This makes it easier to take advantage of new technologies.

APIs can also incorporate multiple types of security measures. They can also provide flexibility with security filtering, providing fine-grained control over which users can access which data.

Services

The data fabric can be built in “blocks,” which make it easier to reuse code. For example, there may be others within your company who do not have a use for the full data fabric, but who do need the code you created to clean the data or to ingest it. You can make that code generic and encapsulate it in a service that anyone can use.

Products

An entire data fabric can be bundled as a product and sold. This would be particularly attractive if made specifically for an industry, such as healthcare.

Why Transition from an MDW to a Data Fabric Architecture?

There are several reasons organizations decide to transition from an MDW to a data fabric. While MDWs have traditionally been the foundation for many businesses’ data infrastructures, they can be rigid and are sometimes hard to scale as data types continue to evolve rapidly. In contrast, a data fabric is designed with scalability in mind. Its inherent flexibility allows it to adapt readily to various data types and sources, making it resilient to current data demands and, to some extent, future-proof.

In addition, the sheer variety of data sources can be overwhelming. A data fabric can weave varied sources together seamlessly, offering a singular, unified view of the data that makes it much easier and more efficient to handle multifaceted data streams.

In a swiftly changing business environment, the ability to process data in real time and derive immediate insights is crucial. A data fabric meets this need by supporting real-time data processing, granting businesses instant access to the latest data.

Furthermore, as concerns about data breaches increase and the emphasis on adhering to data protection regulations intensifies, data security and governance become even more critical. A data fabric addresses these issues with advanced access policies, providing a more secure data environment. It also enhances data governance by restricting access to properly authorized users, protecting sensitive information from potential threats. And for multinational companies that operate across different jurisdictions, each with distinct data protection regulations, a data fabric’s advanced governance capabilities help can maintain compliance no matter where the data is.

Lastly, certain industries—like stock trading and ecommerce—are characterized by rapidly shifting market conditions. In these volatile worlds, staying updated in real time is not just an advantage but a necessity. This immediacy is where a data fabric proves invaluable; its support for real-time data processing keeps such businesses agile, informed, and ready to adapt at a moment's notice.

Potential Drawbacks

While a data fabric architecture has numerous advantages, it isn't devoid of challenges. Transitioning from an MDW to a data fabric can be resource intensive, with initial hiccups related to cost, training, and integration. Also, not all businesses need the advanced capabilities of a data fabric—for small businesses with limited data sources and straightforward processing needs, an MDW might suffice.

Moreover, the inherent complexity of a data fabric can make troubleshooting more challenging. It's crucial to ensure that your organization has the required expertise in-house or accessible through partners before making the shift.

While data fabric offers a cutting-edge solution to modern data management challenges, it's essential for every business to evaluate its unique needs, potential return on investment, and long-term strategic goals before making the transition.

Summary

In this chapter, we delved into the concept of data fabric architecture, an advanced evolution of the MDW designed to enhance data accessibility, security, discoverability, and availability. I defined data fabric and contrasted my definition with industry perspectives, highlighting differences such as the role of data virtualization. The chapter outlined eight key technologies that transition an MDW into data fabric territory, without making a strict demarcation between the two.

I also explained the core components of the data fabric architecture, including data access policies, metadata catalog, master data management, real-time processing, APIs, and optional elements like data virtualization. The chapter emphasizes data fabric's adaptability, especially in handling large data volumes and real-time processing demands.

We looked at reasons for transitioning from an MDW to a data fabric, as well as potential drawbacks such as complexity and resource-intensive transitions. In short, every organization should carefully evaluate its unique needs and capabilities before embracing this cutting-edge solution.

